



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

CS1004-1 – INTRODUCTION TO C PROGRAMMING
 (ACT, AD, AM, CB, CC, CS, EC, EE, IS, RI, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- A Byte consists of**
 A) 1 bit
 B) 2 bits
 C) 4 bits
 D) 8 bits
- Which of the following is not a valid variable name declaration?**
 A) int _a3;
 B) int a_3;
 C) int 3_a;
 D) int _3a;
- Which is correct with respect to the size of the data types?**
 A) char > int > float
 B) int > char > float
 C) char < int < double
 D) double > char > int
- Which of the following shows the correct hierarchy of arithmetic operations in C**
 A) / + * -
 B) * - / +
 C) + - / *
 D) * / + -
- Which of the following is not a binary operator**
 A) Bitwise AND
 B) Logical OR
 C) Bitwise Complement
 D) Modulus
- What will be the output of the following C code?**

```
#include <stdio.h>
int main()
{
  int i = -3;
  int k = i % 2;
  printf("%d\n", k);
  return 0;
}
```

 A) Compile time error
 B) -1
 C) 1
 D) Implementation defined
- What is the output of C Program.?**

```
int main()
{
  int k;

  for(;;)
  {
    printf("TESTING\n");
    break;
  }

  return 0;
}
```

 A) TESTING
 B) No Output
 C) Compiler error
 D) None of the above

8. What is the way to suddenly come out of or Quit any Loop in C Language?

- A) continue statement B) break statement
C) leave statement D) quit statement

9. What is the output of C Program?

```
int main()
```

```
{
```

```
    int a=32;
```

```
    do
```

```
    {
```

```
        printf("%d ", a);
```

```
        a++;
```

```
    }while(a <= 30);
```

```
    return 0;
```

```
}
```

- A) 32
B) 33
C) 30
D) No Output

10. int main()

```
{
```

```
    int a = 10.5;
```

```
    printf("%d",a);
```

```
    return 0;
```

```
}
```

What is the output?

- A) 10.5 B) 10
C) 0 D) Compilation Error

11. A string constant in C is terminated by

- A) '\0' B) '\\0'
C) " D) " "

12. Which function will you choose to join two words?

- A) strlen() B) strcat()
C) strcpy() D) memcat()

13. What is the output of this program?

```
void main()
```

```
{
```

```
    int a[8] = {1,2,3,4,5};
```

```
    printf("%d", a[5]);
```

```
}
```

- A) 5 B) 6
C) 0 D) Garbage value

14. EOF means

- A) End Of File B) End for file
C) File end D) None of the above

15. Which function is used to close opened files?

- A) Open B) fopen
C) Close D) fclose

16. Which of the following is a ternary operator?

- A) && B) >>=
C) ?: D) ->

17. What will be the following code's output if choice = 'R'?
- ```
switch(choice)
{
 case 'R' : printf("RED");
 case 'W' : printf("WHITE");
 case 'B' : printf("BLUE");
 default: printf("ERROR"); break;
}
```
- A) RED  
B) WHITE  
C) RED WHITE BLUE ERROR  
D) ERROR
18. What is the output of C Program with arrays.?
- ```
int main()
{
    char str[25];
    scanf("%s", str);
    printf("%s",str);
    return 0;
}
```
- //input: South Africa
- A) South
B) South Africa
C) S
D) Compiler Error
19. what is the effect of the following code?
- ```
int main()
{
 int a[4]={1,5};
 Printf("%d",a[3]);
}
```
- A) 0  
B) syntax error because of improper initialization  
C) 5  
D) Syntax error because of invalid syntax.
20. What will be the final value of x in the following C code?
- ```
#include <stdio.h>
void main()
{
    int x = 5 * 9 / 3 + 9;
}
```
- A) 3.75
B) Depends on compiler
C) 24
D) 3

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Unit – I		Marks	BT*	CO*	PO*
1.	a) Which are the 4 Fundamental Data types in C? Explain with examples.	8	L*1	1	1
	b) Explain the following with an example:				
	i) Modulus operator				
	ii) Increment operator	4	L2	1	1
	c) Write a C program to find Largest of 3 numbers using If Else Loop.	4	L3	2	1
2.	a) Explain the 2 methods of Type Conversion in C with examples.	6	L2	1	1
	b) What is Ternary operator? Explain its syntax and give examples.	4	L1	2	1
	c) Write a C program to check if a number is Even or Odd and display a suitable message.	6	L3	2	1

- | | | | | | |
|-------|---|---|----|---|---|
| 3. a) | Which are the 2 types of branching using goto statement. Explain with examples. | 8 | L2 | 2 | 1 |
| b) | Write the Syntax of Do-While statement and give an example. | 3 | L1 | 2 | 1 |
| c) | Write a C program to display numbers from 1 to N where N is taken as Input. | 5 | L3 | 2 | 1 |

Unit – II

- | | | | | | |
|-------|---|---|----|---|---|
| 4. a) | What are Strings? Explain any 3 String manipulation functions with examples. | 6 | L1 | 4 | 1 |
| b) | Explain how 2D arrays are declared and initialized. | 4 | L2 | 3 | 1 |
| c) | Write a C program to perform Binary Search on an array of N integers and display suitable messages. | 6 | L3 | 3 | 1 |
| 5. a) | Mention the various categories of User Defined Functions in C. Explain any 2. | 8 | L1 | 3 | 1 |
| b) | Explain how Single dimensional arrays are declared and initialized. | 4 | L2 | 3 | 1 |
| c) | Write a C program to read numbers into a MxN matrix and display the elements. | 4 | L3 | 3 | 1 |
| 6. a) | Which are the 2 Categories of Functions in C. Explain with examples. | 6 | L1 | 3 | 1 |
| b) | Explain the reading of strings during Compile time. | 4 | L2 | 4 | 1 |
| c) | Write a C program to Sort an array of integers using Selection Sort. | 6 | L3 | 3 | 1 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|---|
| 7. a) | With an example, explain the syntax of defining a structure. | 6 | L1 | 5 | 1 |
| b) | Explain how pointer variables are declared with examples. | 4 | L2 | 5 | 1 |
| c) | Write a C program to read N integers into an array A and find the sum of elements using pointers. | 6 | L3 | 5 | 1 |
| 8. a) | Mention and explain any 3 File Functions. | 6 | L1 | 5 | 1 |
| b) | With an example, explain how Files can be Closed. | 4 | L2 | 5 | 1 |
| c) | Write a C program using Structures to store and display the Name, Roll No and Marks of a student. | 6 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
